Content Engine v3: Setup Guide

Get the engine running on a Mac or Linux machine, with scheduled jobs. Takes about 30 minutes. Windows works through WSL (see step 8).

0. Before you start

You need	Where to get it
A Mac or Linux machine that stays on (WSL works for testing)	Only one machine should run the schedule
Python 3.11+, ffmpeg, about 2 GB free disk space	step 2 installs them
Claude Code, signed in to your Claude subscription	claude.com/claude-code
Notion integration token, connected to the Content Engine page	step 3
Optional: free YouTube Data API key; Telegram bot	steps 3 and 9

1. Get the code onto the machine

Copy the whole **engine-v2** folder, or clone it once a GitHub remote exists. Also copy **.env** by hand; it is never in git.

```
cd ~/content-engine        # the folder you copied
ls cli.py                  # check you are in the right place
```

2. Install

This one-time setup works on macOS (Homebrew) and Ubuntu/Debian. It is safe to re-run, and doing so updates yt-dlp.

```
scripts/setup.sh
```

It installs ffmpeg, Python, the **.venv** environment, yt-dlp, Whisper and the git hooks. It creates **.env** from **.env.example** and runs the tests. Whisper's first install downloads 1–2 GB.

3. Fill in .env

Key	Needed?	What it is
`NOTION_TOKEN`	Required	Notion → Settings → Connections → Develop integrations → copy the secret. Then on the Content Engine page: ••• → Connections → add the integration.
`CLAUDE_CODE_OAUTH_TOKEN`	Servers	Paste the token from step 4.
`YOUTUBE_API_KEY`	Recommended	Free. Google Cloud → YouTube Data API v3 → Credentials. Gives faster, steadier creator scans.
`ENGINE_COOKIES_FILE`	Servers	Path to cookies.txt (see step 6).
`TELEGRAM_BOT_TOKEN` `TELEGRAM_CHAT_ID`	Optional	Notifications (see step 9).

4. Sign in to Claude

```
claude            # desktop: run once, then type /login
claude setup-token   # server: copy the token into .env as CLAUDE_CODE_OAUTH_TOKEN
```

All AI steps run through your Claude subscription. No API key is needed and there is no per-call bill.

5. Check everything connects

```
.venv/bin/python cli.py ping --claude     # every line should say OK
.venv/bin/python cli.py settings-check    # shows the settings in use
```

Notion 401 means the token is wrong. **Notion 404** means the integration is not connected to the Content Engine page.

6. Logins for Instagram (and YouTube or TikTok when they ask)

Machine	What to do
Desktop	Log in to Instagram and YouTube in Firefox on that machine. The engine reads that login. A secondary account is safer.
Server	Export cookies.txt from a logged-in browser (any cookies.txt extension works), copy it to the server and set `ENGINE_COOKIES_FILE=/path/cookies.txt`. Refresh it every few weeks, when links start coming back as Needs Input.

7. Set the time zone

Schedule times are in the machine's local time. Set it to Asia/Karachi, or shift the hours in `scheduling/jobs.txt`.

```
sudo timedatectl set-timezone Asia/Karachi    # Linux
# Mac: System Settings → General → Date & Time
```

8. Turn on the schedule

```
scheduling/install.sh                              # preview only
scheduling/install.sh --apply --dry-run-commands   # rehearsal: no Notion writes
scheduling/install.sh --apply                      # the real thing
loginctl enable-linger $USER                       # Linux: run while logged out
```

Job	When	What it does
reschedule	Daily 07:30	Acts on your Approved / Postponed / Rejected choices
news-radar	Daily 08:00	Adds up to 5 news items as Pending
add-url	Daily 09:00	Processes links you added in Notion and transcripts you pasted
generate-cards	Daily 09:30	Writes cards (A/B) for locked topics, plus revisions
scan-creators	Thu 20:00	Scans Active creators for topics that are doing well
weekly-plan	Fri 17:00	Fills and locks next week's slots

This uses systemd on Linux and launchd on Mac. Use `scheduling/install.sh cron --apply` for cron instead. **Windows:** `powershell -ExecutionPolicy Bypass -File scheduling\windows\register-tasks.ps1 -Apply` (the PC must stay on). **Stop it:** `scheduling/install.sh --remove --apply`.

9. Notifications (optional)

Create a bot: message **@BotFather** → /newbot → put the token in .env. Send your bot any message, then open api.telegram.org/bot<token>/getUpdates and copy chat.id into .env.

```
.venv/bin/python cli.py settings-set "Notifications" On --apply
.venv/bin/python cli.py notify-test --apply
```

10. First-run checklist in Notion

Do this	Why
Content Creators: set State = Active on the creators to scan	Only Active creators are scanned
Run `cli.py score --apply` once	Ideas need a brand-fit score to fill a slot
News Candidates: set each item to Approved or Rejected	Only approved news becomes an idea
Brand DNA page: fill in Topics to avoid and the good and bad idea examples	Sharpens brand-fit judging

11. Day-to-day: how to tell it's working

Where	What you see
Notion → Run Log	One row for every scheduled run that wrote something: status, counts, errors
`logs/cron/<job>.log`	The full output of each scheduled job
`logs/changeset-*.json`	Every change a run planned or made
Any command without --apply	A preview only. Nothing is written to Notion

12. Quick fixes

Problem	Fix
Claude Code isn't signed in	Run claude → /login, or set CLAUDE_CODE_OAUTH_TOKEN
YouTube: "confirm you're not a bot"	Log in to YouTube in Firefox, or refresh cookies.txt
All Instagram links fail	Cookies expired: refresh the login or cookies.txt
KeyError: 'news_candidates' (or another database)	`cli.py migrate-schema --apply`
A run changed something wrongly	Open Notion page history to restore the row. The changeset in logs/ shows what changed
Downloads break after a site update	Re-run `scripts/setup.sh` (it updates yt-dlp)

More detail: docs/operations.md · Commands: README.md · Progress: PROGRESS.md